

FIT5032 Assessed Lab 4

Yilian Sun

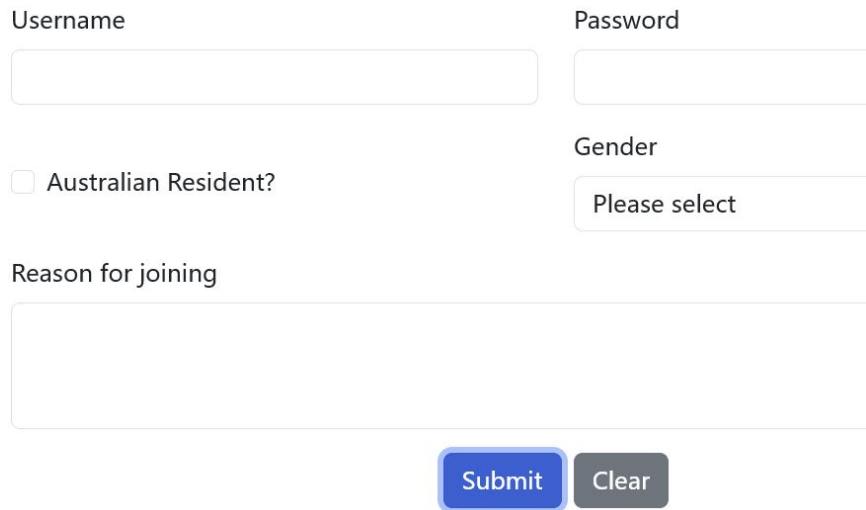
Student ID: [Student ID to be completed]

This PDF contains the real implementation evidence, test outcomes, and Git history for the Week 3 Library Vue project extended for Assessed Lab 4.

eFolio Task 4.1

The original form, HTML built-in validation stage, and final Vue custom validation stage were implemented and captured below.

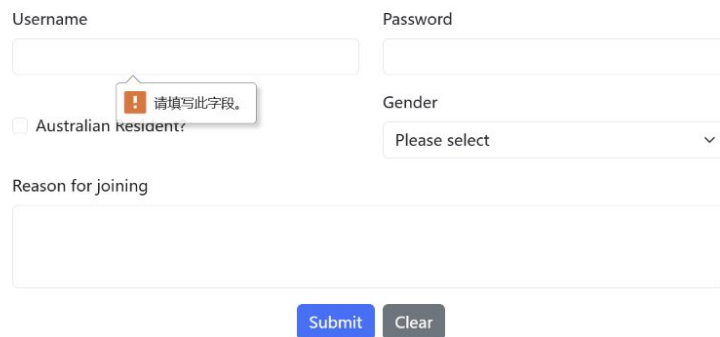
User Information Form



A screenshot of a web form titled "User Information Form". It contains the following fields: "Username" (text input), "Password" (text input), "Australian Resident?" (checkbox), "Gender" (dropdown menu with "Please select" text), and "Reason for joining" (text area). At the bottom right are "Submit" and "Clear" buttons. The form is shown in a state where it can be submitted without any data.

Figure 1. Original Week 3 form allows an empty submission and creates an empty card.

User Information Form



A screenshot of the same "User Information Form" as in Figure 1, but with HTML built-in validation. The "Username" field is empty and has a red error message box above it that says "请填写此字段。" (Please fill in this field). The "Submit" button is disabled (greyed out), while the "Clear" button remains active. The other fields and the form structure are identical to Figure 1.

Figure 2. HTML built-in validation prevents submission when Username is empty.

User Information Form

Username

Password

☐ Australian Resident?

Gender

Please select
▼

Reason for joining

Submit

Clear



Figure 3. HTML built-in validation evidence for password length checking with password value 123.

User Information Form

Username

!

Username is required

Password

Password must be at least

Australian Resident?

Please select
!
▼

Please choose Yes or No

Gender

Please select
▼

Please select a gender

Reason for joining

Reason for joining is required

Figure 4. Vue validation prevents a missing username from being submitted.

User Information Form

Username Yilian	Password Password must include an
Australian Resident? Please select   Please choose Yes or No	Gender Please select Please select a gender
Reason for joining Reason for joining is required	

Figure 5. Vue password validation reports an invalid password and blocks submission.

User Information Form

Username Yilian	Password Password must be at least
Australian Resident? Please select   Please choose Yes or No	Gender Please select Please select a gender
Reason for joining short Reason must be at least 10 characters	

Figure 6. Vue validation displays additional Resident, Gender, and Reason errors.

eFolio Task 4.2

Field	Final validation rule
Username	Trimmed; required; at least 3 characters; spaces-only input rejected.
Password	At least 8 characters; includes uppercase, lowercase, number, and special character.
Australian Resident	User must explicitly choose Yes or No from a blank default select.
Gender	User must explicitly choose Male, Female, or Other from a blank default select.
Reason	Trimmed; required; 10 to 200 characters.

User Information Form

Username

Password

Australian Resident?

Gender

Reason for joining

User Information
Username: Yilian
Password: Valid123!

Figure 7. A valid record is accepted and displayed after passing all five validations.

User Information Form

Username	Password
Australian Resident?	Gender
Please select	Please select
Reason for joining	
Submit Clear	

User Information
Username: Yilian
Password: Valid123!
Australian Resident: Yes
Gender: Female
Reason: I want to discover new books.

User Information
Username: Alex
Password: ReadMore2@
Australian Resident: No
Gender: Male
Reason: I enjoy reading with the library community.

User Information
Username: Taylor
Password: Library3#

Figure 8. PrimeVue DataTable displays three valid submitted user records.

```

Form.vue - Five Vue validation functions
01 const validateName = (blur) => {
02   const username = formData.value.username.trim()
03
04   if (username) {
05     if (blur) errors.value.username = 'Username is required'
06     return false
07   }
08
09   if (username.length < 3) {
10     if (blur) errors.value.username = 'Username must be at least 3 characters'
11     return false
12   }
13
14   errors.value.username = null
15   return true
16 }
17
18 const validatePassword = (blur) => {
19   const password = formData.value.password
20   const hasUppercase = /[A-Z]/.test(password)
21   const hasLowercase = /[a-z]/.test(password)
22   const hasNumber = /[0-9]/.test(password)
23   const hasSpecialChar = /[!@#$%^&*()_+~`{|}\<.>:;"/.test(password)
24
25   if (password.length < 8) {
26     if (blur) errors.value.password = 'Password must be at least 8 characters'
27     return false
28   }
29
30   if (!hasUppercase) {
31     if (blur) errors.value.password = 'Password must include an uppercase letter'
32     return false
33   }
34
35   if (!hasLowercase) {
36     if (blur) errors.value.password = 'Password must include a lowercase letter'
37     return false
38   }
39
40   if (!hasNumber) {
41     if (blur) errors.value.password = 'Password must include a number'
42     return false
43   }
44
45   if (!hasSpecialChar) {
46     if (blur) errors.value.password = 'Password must include a special character'
47     return false
48   }
49
50   errors.value.password = null
51   return true
52 }
53
54 const validateResident = (blur) => {
55   if (!formData.value.resident) {
56     if (blur) errors.value.resident = 'Please choose Yes or No'
57     return false
58   }
59
60   errors.value.resident = null
61   return true
62 }
63
64 const validateGender = (blur) => {
65   if (!formData.value.gender) {
66     if (blur) errors.value.gender = 'Please select a gender'
67     return false
68   }
69
70   errors.value.gender = null
71   return true
72 }
73
74 const validateReason = (blur) => {
75   const reason = formData.value.reason.trim()
76
77   if (!reason) {
78     if (blur) errors.value.reason = 'Reason for joining is required'
79     return false
80   }
81
82   if (reason.length < 10) {
83     if (blur) errors.value.reason = 'Reason must be at least 10 characters'
84     return false
85   }
86
87   if (reason.length > 200) {
88     if (blur) errors.value.reason = 'Reason must be 200 characters or fewer'
89     return false
90   }
91
92   errors.value.reason = null
93   return true
94 }
95
96 const validateForm

```

Figure 9. Source evidence for the five Vue validation functions.

Form.vue - PrimeVue DataTable template and imports

```
01  import DataTable from 'primevue/datatable'
02  import Column from 'primevue/column'
03
04  <section class="datatable-section mt-5" v-if="submittedCards.length">
05      <h2 class="h4 mb-3">Submitted User Information</h2>
06      <DataTable
07          :value="submittedCards"
08          stripedRows
09          showGridlines
10          responsiveLayout="scroll"
11          class="p-datatable-sm"
12      >
13          <Column field="username" header="Username" sortable />
14          <Column field="password" header="Password" />
15          <Column header="Australian Resident">
16              <template #body="{ data }">
17                  {{ formatResident(data.resident) }}
18              </template>
19          </Column>
20          <Column header="Gender">
21              <template #body="{ data }">
22                  {{ formatGender(data.gender) }}
23              </template>
24          </Column>
25          <Column field="reason" header="Reason" />
26      </DataTable>
27  </section>
```

Figure 10. Source evidence for PrimeVue DataTable imports and template.

main.js - PrimeVue Aura configuration

```
01  import { createApp } from 'vue'
02  import App from './App.vue'
03  import PrimeVue from 'primevue/config'
04  import Aura from '@primevue/themes/aura'
05
06  import 'bootstrap/dist/css/bootstrap.min.css'
07  import 'bootstrap/dist/js/bootstrap.bundle.min.js'
08
09  const app = createApp(App)
10
11  app.use(PrimeVue, {
12    theme: {
13      preset: Aura
14    }
15  })
16
17  app.mount('#app')
```

Figure 11. Source evidence for PrimeVue Aura configuration in main.js.

Git commit history

```
10f2024 (HEAD -> main) lab4: add PrimeVue DataTable
a77344b lab4: implement five Vue form validations
2b66601 lab4: add HTML built-in form validation
352e1cc (origin/main) Complete Lab 3 Bootstrap responsive form
56e07d8 Fix bookstore data display
9ec0e26 Use official lab 2 files and complete activities
94c2f9e Complete assessed lab 2
```

Figure 12. Git commit history showing staged assessed lab updates.

Testing Summary

Scenario	Result	Evidence
Original empty submission creates an empty card before validation	Pass	Figure 1
HTML required username blocks empty submit	Pass	Figure 2
HTML password length rule added and checked with value 123	Pass	Figure 3
Vue empty/short username blocks submit	Pass	Figure 4
Vue password complexity blocks invalid passwords	Pass	Figure 5
Resident, Gender, and Reason extra validations block submit	Pass	Figure 6
Valid data submits successfully	Pass	Figure 7
Three valid submissions display as three DataTable rows	Pass	Figure 8
Browser console errors during DataTable capture	Pass - none observed	Browser automation log
npm run build	Pass	Vite production build completed successfully

Build and Git Summary

Build result: npm run build completed successfully after PrimeVue integration.

Student action still required: fill in the Student ID placeholder, confirm tutor sharing on the private GitHub repository, and upload this PDF to Moodle.